

MIAMI NHC, FL, USA

WMO: 722020

Lat: 25.7552N

Lon: 80.3836W

Elev: 29

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.0	52.7	32.1	26.6	55.7	37.0	32.3	57.4	21.8	75.1	20.1	73.7	7.2	330	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	92.0	77.7	90.9	77.7	90.0	77.6	80.4	86.8	79.9	86.7	79.3	86.4	9.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	148.9	83.6	77.9	145.3	83.4	77.3	142.1	83.2	43.9	86.8	43.2	86.6	42.6	86.4	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	18.1	16.5	42.8	94.8	4.0	1.7	39.9	96.0	37.6	97.0	35.3	97.9	32.4	99.1		
(5)			37.6	81.9	4.5	1.3	34.3	82.8	31.7	83.6	29.1	84.4	25.8	85.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.5	68.4	70.7	73.0	76.7	80.2	82.7	84.0	84.1	83.0	80.0	74.7	71.4
	DBStd	6.96	6.91	6.18	5.34	3.74	2.69	2.43	2.10	2.07	2.17	3.50	4.58	6.46
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	112	49	24	10	0	0	0	0	0	0	0	3	26
	CDD50	10022	572	581	713	800	938	982	1054	1058	989	931	740	664
	CDD65	4660	156	185	258	350	473	532	589	593	539	466	294	225
	CDH74	43230	612	855	1445	2500	4421	5836	7058	7082	6017	4442	1830	1132
	CDH80	13954	37	94	251	535	1287	2150	2940	2928	2180	1204	249	99
(14) Wind	WSAvg	7.9	8.1	8.2	9.2	9.5	8.7	7.1	6.8	6.6	6.9	8.2	8.1	7.9
(15) Precipitation	PrecAvg	59.6	1.9	2.1	2.3	3.1	5.9	9.4	6.1	7.8	9.1	6.9	3.1	2.0
	PrecMax	87.0	7.6	6.8	6.6	10.6	18.5	22.4	12.8	16.7	24.4	21.7	13.9	9.8
	PrecMin	33.5	0.2	0.0	0.2	0.1	0.4	2.0	1.7	2.3	1.8	0.7	0.4	0.1
	PrecStd	12.0	1.7	1.7	1.7	2.6	4.3	4.8	2.6	3.6	4.4	4.7	2.5	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.1	85.0	87.9	89.2	91.7	93.0	93.2	93.2	92.3	90.2	86.4	84.3
		MCWB	72.4	71.8	73.4	72.9	75.2	77.5	77.8	78.0	77.7	77.4	75.0	74.2
	2%	DB	80.9	82.5	85.1	86.8	89.2	91.1	91.5	91.6	90.5	88.3	84.4	82.4
		MCWB	71.4	71.8	72.8	73.2	74.9	77.5	77.9	78.2	77.7	76.9	74.3	73.1
	5%	DB	79.3	81.0	82.8	84.9	87.5	89.6	90.4	90.5	89.3	86.9	82.8	81.0
		MCWB	70.3	71.2	71.8	72.5	74.6	77.3	77.8	78.2	77.7	76.3	73.3	72.4
	10%	DB	77.6	79.2	80.7	83.2	86.1	88.2	89.3	89.4	88.0	85.5	81.2	79.5
		MCWB	69.5	70.2	70.5	71.7	74.1	77.0	77.7	78.2	77.5	75.6	72.2	71.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.2	75.5	77.3	77.0	79.2	80.6	80.5	81.0	81.3	80.9	78.1	76.9
		MCDB	79.0	79.5	82.3	83.6	85.4	87.0	88.1	88.1	86.6	84.7	82.0	80.6
	2%	WB	73.5	74.1	75.3	75.7	77.7	79.6	80.0	80.2	80.3	79.3	76.7	75.6
		MCDB	78.0	78.7	81.2	82.1	84.5	86.8	87.8	87.5	86.1	84.4	81.0	79.7
	5%	WB	72.4	73.0	73.9	74.8	76.8	78.8	79.3	79.6	79.5	78.3	75.6	74.6
		MCDB	77.1	78.0	79.6	81.3	83.7	86.0	87.2	87.1	85.8	83.8	80.3	78.8
	10%	WB	71.1	71.8	72.7	73.7	76.0	78.1	78.5	79.1	78.8	77.5	74.6	73.3
		MCDB	76.0	77.3	78.6	80.5	82.8	85.4	86.3	86.5	85.0	83.4	79.5	77.6
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	13.8	13.7	12.7	12.1	11.7	11.6	11.6	11.0	10.5	11.9	12.5
		MCDBR	12.5	13.3	14.4	14.0	13.4	12.7	12.3	12.3	11.7	11.0	11.3	11.3
		MCWBR	6.2	5.6	6.1	5.7	5.2	5.1	4.9	4.7	4.4	4.1	5.1	5.5
	5% WB	MCDBR	11.8	11.7	13.3	12.5	12.5	12.3	11.7	11.6	11.1	10.2	10.3	10.3
		MCWBR	7.0	5.9	6.7	6.2	5.5	5.7	5.3	5.1	4.9	4.5	5.4	6.3
(40) Clear-Sky Solar Irradiance	taub		0.364	0.375	0.388	0.398	0.426	0.458	0.463	0.455	0.432	0.415	0.381	0.373
	taud		2.524	2.503	2.456	2.414	2.381	2.348	2.327	2.378	2.440	2.454	2.497	2.516
	Ebn at Noon		282	287	288	286	276	265	264	266	271	270	274	274
	Edh at Noon		30	32	36	38	39	40	41	39	36	34	30	29
(44) All-Sky Solar Radiation	RadAvg		1175	1419	1735	1961	1965	1783	1833	1768	1559	1441	1240	1090
	RadStd		75	79	80	91	158	156	79	94	78	99	57	81

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.58	+1.39	N/A	N/A	N/A	N/A	-29	+205	+167
(47) Regional (29 neighbors)		+0.68	+1.58	N/A	N/A	N/A	N/A	-32	+236	+200

Nomenclature: See separate page